COHERENT SHEAVES WITH FINITE SUPPORT ON A PROPER SCHEME ARE GLOBALLY GENERATED, ACYCLIC, AND HAVE EULER CHARACTERISTIC LINEAR IN THE RANK OF A LOCALLY FREE TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Cohomology of coherent sheaves on affine schemes and on relative affine morphisms. Stacks Project, Cohomology of Schemes, Lemmas coherent-lemma-quasi-coherent-affine-cohomology-zero (tag 01XB) and coherent-lemma-relative-affine-cohomology (tag 089W). Gives global generation and the vanishing of higher cohomology on the affine support, and the agreement of the cohomologies of F and G.
- Zero-dimensional locally Noetherian schemes are affine, and algebraic schemes of dimension zero are finite disjoint unions of spectra of local Artinian rings. Stacks Project, Properties, Lemma properties-lemma-locally-Noetherian-dimension-0 (tag 0AAX) and Varieties, Lemma lemma-algebraic-scheme-dim-0 (tag 06LH). Supplies affineness of Z and the Artinian decomposition used to write H^0 as a direct sum of stalks.
- Scheme-theoretic support of a coherent sheaf and the projection formula. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-coherent-support-closed (tag 01Y5) and Cohomology, Lemma cohomology-lemma-projection-formula (tag 01E8). Produces the closed immersion i : Z -> X and computes F tensor E through i_*.

Source reference. Original reference 01Y5: sources/stacks-chow/tags.txt, source lines 2544–2544. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AAX: sources/stacks-chow/tags.txt, source lines 13000–13000. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original lemma 06LH: sources/stacks-chow/tags.txt, source lines 8466–8466. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{F}$ be a coherent sheaf with $\dim(Supp(\mathcal{F})) \leq 0$. Then*

(1) *$\mathcal{F}$ is generated by global sections,*
(2) *$H^0(X, \mathcal{F}) = \bigoplus_{x \in Supp(\mathcal{F})} \mathcal{F}_x$,*
(3) *$H^i(X, \mathcal{F}) = 0$ for $i > 0$,*
(4) *$\chi(X, \mathcal{F}) = \dim_k H^0(X, \mathcal{F})$, and*
(5) *$\chi(X, \mathcal{F} \otimes \mathcal{E}) = n\chi(X, \mathcal{F})$ for every locally free module $\mathcal{E}$ of rank n.*

Proof. By Cohomology of Schemes, Lemma 01Y5 we see that $\mathcal{F} = i_*\mathcal{G}$ where $i : Z \to X$ is the inclusion of the scheme theoretic support of $\mathcal{F}$ and where $\mathcal{G}$ is a coherent $\mathcal{O}_Z$-module. By definition of the scheme theoretic support the underlying topological space of Z is $\mathrm{Supp}(\mathcal{F})$. Since the dimension of Z is 0, we see Z is affine (Properties, Lemma 0AAX). Hence $\mathcal{G}$ is globally generated and the higher cohomology groups of $\mathcal{G}$ are zero (Cohomology of Schemes, Lemma 01XB). In fact, by Lemma 06LH the scheme Z is a finite disjoint union of spectra of local Artinian rings. Thus correspondingly $H^0(Z, \mathcal{G}) = \bigoplus_{z \in Z} \mathcal{G}_z$. The cohomologies of $\mathcal{F}$ and $\mathcal{G}$ agree by Cohomology of Schemes, Lemma 089W. Thus $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $H^0(X, \mathcal{F}) = H^0(Z, \mathcal{G})$. In particular we have (3) is true. For $z \in Z$ corresponding to $x \in \mathrm{Supp}(\mathcal{F})$ we have $\mathcal{G}_z = (i_*\mathcal{G})_x = \mathcal{F}_x$. We conclude that (2) holds. Of course (2) implies (1). We have (4) by definition of the Euler characteristic $\chi(X, \mathcal{F})$ and (3). By the projection formula (Cohomology, Lemma 01E8) we have

$$i_*(\mathcal{G} \otimes i^*\mathcal{E}) = \mathcal{F} \otimes \mathcal{E}.$$

Since Z has dimension 0 the locally free sheaf $i^*\mathcal{E}$ is isomorphic to $\mathcal{O}_Z^{\oplus n}$ and arguing as above we see that (5) holds. □